

COMP4431 Artificial Intelligence

Quiz 04

Suggested Solution

1.

(a) [2.5 marks]

Step 1: Calculate the Entropy of the Target Attribute (Go Hiking)

The target attribute "Go Hiking" has the following distribution:

- Yes: 4
- No: 4

The entropy $H(\text{whole_table})$ is calculated as:

$$H(\text{whole_table}) = -((4/8)\log(4/8) + (4/8)\log(4/8)) = 1$$

Step 2: Calculate the Entropy for Attribute Temperature

- **C_Hot:** ID {1,2,3} -> Go_Hiking { No, No, Yes}
 - Entropy = $-((2/3)\log(2/3) + (1/3)\log(1/3)) \approx 0.918$
- **C_Mild:** ID {4,8} -> Go_Hiking { No, Yes}
 - Entropy = $-((1/2)\log(1/2) + (1/2)\log(1/2)) = 1$
- **C_Cool:** ID {5,6,7} -> Go_Hiking { Yes, No, Yes}
 - Entropy = $-((2/3)\log(2/3) + (1/3)\log(1/3)) \approx 0.918$

Weighted Entropy for Temperature:

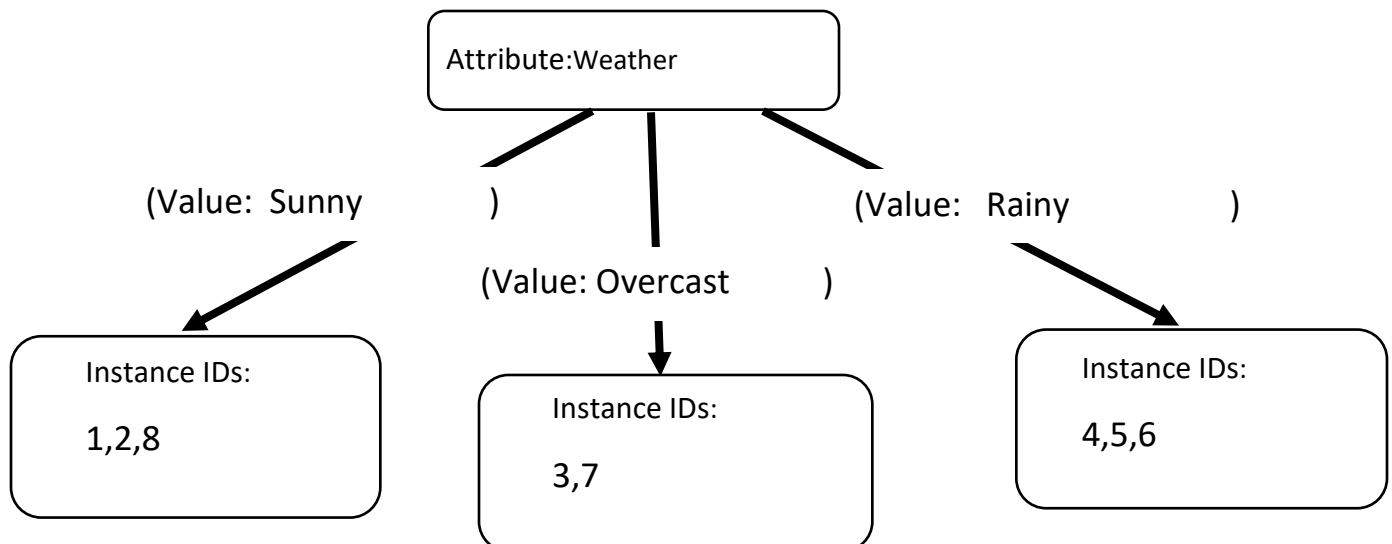
$$H(\text{Temperature}) = 3/8 \times 0.918 + 2/8 \times 1 + 3/8 \times 0.918 \approx 0.9385$$

Information Gain for Temperature:

$$IG(\text{Temperature}) = 1 - 0.9385 = 0.0615$$

(b) [1 mark]

The attribute with the highest information gain is "Weather" with an information gain of 0.656. Therefore, "Weather" should be chosen as the root node.



2.

a) [2.5 marks]

Step1. Calculate the weighted sum (z):

The weighted sum z is calculated using the formula:

$$s = w_1 \cdot x_1 + w_2 \cdot x_2 + b$$

Given:

- $w_1 = 0.5$
- $w_2 = -0.3$
- $b = 0.1$
- $x_1 = 0.8$
- $x_2 = 0.4$

Substitute these values into the formula:

$$\begin{aligned} s &= (0.5 \times 0.8) + (-0.3 \times 0.4) + 0.1 \\ &= 0.4 - 0.12 + 0.1 = 0.38 \end{aligned}$$

Step2. Apply the sigmoid activation function:

The sigmoid function $\sigma(x)$ is defined as:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Substitute $s = 0.38$ into the sigmoid function:

$$\sigma(0.38) = \frac{1}{1 + e^{-0.38}}$$

$$\sigma(0.38) = 1 / (1 + 0.684) \approx 1 / 1.684 \approx 0.593$$

Calculate $e^{-0.38}$

$$e^{-0.38} \approx 0.684$$

Therefore:

$$\sigma(0.38) = 1 / (1 + 0.684) = 1 / 1.684 \approx 0.594$$

Thus, the output of the neuron after applying the sigmoid activation function is approximately **0.594**.

b) [1 mark]

The sign function is not suitable for use in neural networks during backpropagation because:

(**either one of the following, or other acceptable answers)

Non-Differentiability: The sign function is not differentiable at zero and has a derivative of zero elsewhere, which prevents the calculation of gradients needed for weight updates.

Lack of Gradient Information: With a zero gradient, the network cannot learn effectively, as there is no information on how to adjust the weights to minimize the error.